

Charles “Chaz” Merritt

Curriculum Vitae

Athens, GA · charles.merritt@uga.edu · [LinkedIn](#) · [GitHub](#) · [Website](#)

Research Interests

Reinforcement Learning, Markov Chains, Computational Forestry, Simulation-Based Planning

Education

M.S. in Artificial Intelligence

Expected 2026 — GPA: 3.66

University of Georgia, Institute for Artificial Intelligence

Thesis: *Quantifying Economic Value-at-Risk in Timber Management via Markov Chain Monte Carlo*

Advisor: Pete Bettinger

B.A. in Cognitive Science

2024 — GPA: 3.53

University of Georgia, Institute for Artificial Intelligence

Minor in Computer Science

Research Experience

Research Assistant

Aug. 2024 – May 2026

University of Georgia, Institute for Artificial Intelligence

Applied MCMC and Value-at-Risk (VaR, CVaR) methods for economic risk modeling. Built a web-based interface (Django, TypeScript) for optimization workflows. Co-authored peer-reviewed research at the intersection of AI and forestry.

Other Experience

Geospatial Analyst Intern

May – Sep. 2024

Northwest Management

Designed automated GIS pipelines using Python, Rasterio, and GeoPandas. Implemented clustering algorithms and image segmentation models (SAM) for forest analysis. Processed and analyzed large-scale spatial datasets for operational forestry decision-making.

OMNI Intern

May – Dec. 2023

Oak Ridge National Lab

Supported development of intrusion prevention systems for enterprise networks. Gained hands-on experience with SmartNICs (Nvidia BlueField DPUs). Presented to stakeholders at a Department of Energy poster session in Washington, D.C.

Publications

Peer-Reviewed Articles

Merry, K., Bettinger, P., **Merritt, C.**, Rasheed, K., Lowe III, R. C., & da Silva, B. (2026). [AI Technology Use in North American Forestry](#). *Journal of Forestry*. doi:10.1007/s44392-026-00079-8

Presentations

Merritt, C. (2025). Optimizing Timber Stands under Uncertainty. *Society of American Foresters Conference*.

Merritt, C. (2025). Harnessing Reinforcement Learning for Forest Growth Modeling Under Uncertainty. *AI For Eastern US Forestry Conference*.

Merritt, C. (2023). Online Learning in Applied Forestry. *PERSEUS Annual Meeting*.

Merritt, C. (2023). Rapid & Scalable Forest Inventory with Deep Neural Nets. *Southern Forestry & GIS*

Conference.
Merritt, C. (2023). Exploring AI Frameworks for Next-Generation Cybersecurity with Nvidia Morpheus & SLIPS. *Dept. of Energy OMNI Fire Poster Session.*

Selected Projects

- **Optimization Software Web Interface** (2026). Developed an interactive webapp for the SETS timber supply models. Tools: Django, Python, TypeScript, HiGHS-Py.
- **Brandon AI** (2026). Developed contrastive representation models for phonetic similarity learning with InfoNCE loss. Tools: PyTorch, NumPy.

Honors and Awards

ISACA Foundation Digital Trust Scholarship Jan. 2024

Service

- Club Archery @ UGA, *President* Aug. 2024 – May 2025

Skills

- **Programming:** Python, TypeScript
- **Machine Learning:** PyTorch, Reinforcement Learning, Contrastive Learning, MCMC
- **Modeling:** Stochastic Optimization, Simulation, Value-at-Risk (VaR, CVaR)
- **Data & GIS:** GeoPandas, Spatial Analysis, Clustering, Remote Sensing
- **Tools:** Django, Node.js, HiGHS, NumPy, Railway

Certifications

New Media, University of Georgia May 2024
Applied Data Science, University of Georgia May 2024